

AstroMesh — narration

Spoken text only. Roughly two and a half minutes. Every claim below is visible on screen when it is said.

[Boot / globe]

AstroMesh screens thirty-one thousand tracked objects against the real catalogue for collisions.

Watch the top left. It says blocked. Not error — blocked. Something is outstanding, and nothing gets authorised until it is resolved.

Every irreversible action here passes a gate, and the gate answers in four states: complete, partial, blocked, unresolved.

[Risk tab → VOTE]

A real conjunction, put to the operator vote.

Refused — and the reasons are named. Flight Rule seven: at least one object must be commandable. Both of these are debris. Nobody can move either one.

That rule is non-negotiable. There is no waiver path in the engine — not a permission we withheld, an absence of code.

And these grey rows are not failures. They are unresolved: rules we could not evaluate. Most systems read missing data as no problem found. We treat it as blocking, because we don't know is a different answer from it's fine.

[How tab → corridor draws]

Long March 5B, November 2022. A twenty-tonne stage that came down uncontrolled and closed airspace over Spain and southern France.

Every step names a number and its source.

The corridor is this wide because of the live geomagnetic index. Density uncertainty, plus or minus forty-nine percent — measured, not assumed.

What is underneath comes from DINOv3, a satellite vision model: eighteen million cells at three-point-four kilometres. Validated against a population raster it never saw. Above sixty north it refuses to classify, because we measured that it fails there.

Casualty expectancy: twenty times over the legal limit.

And this line matters. The regulator's own method flattens population into latitude bands — it cannot see which longitude the debris is heading for. Same footprint, roughly half the risk. Across eighteen

thousand footprints, it understates exposure ninety-four percent of the time over dense ground.

[Follow descent]

The same descent, from the vehicle. Altitude, flight path angle, downrange.

The ground scrolling underneath is the vision model — red is built up, blue is water, grey is where it declines to answer.

The corridor and entry parameters are the recorded event. The motion is a kinematic replay, and the caption says so.

[Storms tab → Gannon]

May 2024, the Gannon storm. Half of everything in low Earth orbit manoeuvred within days.

Radio dead over both poles. Navigation degraded inside the auroral ovals. Scintillation at the equator. Our ground stations drop from eight to three.

And that reaches into the rulebook. Flight Rule nineteen needs the command uplink to survive. The Sun just shortened the window to command a satellite, so the deadline to decide moved.

A constraint is not a checkbox. It is something the physical world can take away from you while you are deciding.

[Prove it]

The challenge asked for this to stay theme-independent.

Same engine, zero changes, running a software release gate — tests, CVEs, rollback plans, error budgets. Four states, same precedence.

The rules are data. Orbit is where we needed it first.

[GPU screen → close on the signal]

That just screened every pair in the catalogue — about half a billion — as a compute shader on this laptop's GPU. It is a coarse filter; candidates are re-screened with full propagation. The verdict never comes from it.

Four states. Complete, partial, blocked, unresolved.

The repository carries the proof: a model checker over a hundred and forty-eight thousand states, property tests across thousands of generated rulebooks, and coverage theorems for the bounds.

Two of those results are about what we cannot do.